

Charles Deering McCormick Distinguished
Assistant Professor of Instruction
Northwestern University
Department of Computer Science
Computer Science + Learning Sciences Program
Affiliated Faculty Member
Faculty Chair, Willard Residential College

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Technological Institute
2145 Sheridan Rd., Rm. L374
Evanston, IL 60208

EDUCATION

Northwestern University, Evanston, Illinois

Ph.D. in Computer Science and Learning Sciences, September 2021

Advisor: Dr. Uri Wilensky

Dissertation: Empowering Teachers to CT-ify the Science Classroom

M.S. in Computer Science, September 2021

Thesis: Teaching CS: Identifying Unique Aspects of the Role of a CS Teacher in the Classroom

University of South Carolina Honors College, Columbia, South Carolina

B.S.C.S. in Computer Science, May 2015, *summa cum laude*

B.S. in Mathematics, Minor in Music, May 2015, *summa cum laude*

with Leadership Distinction in Civic & Professional Engagement

RESEARCH INTERESTS

My research lies at the intersection of computer science and the learning sciences and focuses on 1. cognitive studies of computer science learners; 2. teacher learning and practice (specifically pedagogical content knowledge); 3. agent-based modeling in educational contexts. I often advise undergraduate, masters, and doctoral research projects adjacent to these areas in addition to those focused on computer science education, teacher professional development, and AI in education.

Keywords: Computational Thinking & Computational Literacy; Computing Pedagogy; Broadening Participation in Computing; Learning Analytics; Teacher Education; Computer Science Communication

HONORS AND AWARDS

Charles Deering McCormick Distinguished Professor of Instruction (University Teaching Award), 2026

NUCS - Excellence in Teaching Award, 2024 - 2025

NUIT Canvas Hall of Fame - Most Innovative Course Site, 2024 for COMP_SCI 110

NUIT Canvas Hall of Fame - Best Use of Video (Finalist), 2023 for COMP_SCI 111

Associated Student Government Faculty Honor Roll, 2021-2022

McCormick School of Engineering George H. and D. G. Martin Fellowship, Fall 2020

Presidential Scholar Finalist, Northwestern University, Spring 2018

National Science Foundation Graduate Research Fellowship, Spring 2015

Algernon Sydney Sullivan Award, Spring 2015

University of South Carolina's highest honor awarded to two graduating students each year

Barry M. Goldwater Scholar, Spring 2014

Phi Beta Kappa, Spring 2014

Udall Scholar Honorable Mention, Spring 2013

Carolina Leadership Initiative Award, 2012 - 2013

Expanding Scientific Leadership through Carolina Science Outreach; Advisor: Dr. Munn Sanchez

TEACHING (*Course Evaluations Available by Request*)

GEN_ENG 151 - *Honors Programming for Engineers*

Course Designer & Instructor - Northwestern University

F26

GEN_ENG 150 - *Programming for Engineers* **W26 & S26**
 Instructor - Northwestern University

COMP_SCI 110 - *Intro to Computer Programming in Python* **W+S22, W+S23, W+S24, W25, W26**
 Instructor - Northwestern University - Avg. Class Size: 195

COMP_SCI 111 - *Fundamentals of Computer Programming I* **F21, F22, F23, F24, F25, SU26**
 Instructor - Northwestern University - Avg. Class Size: 396

COMP_SCI 372,472 - *Designing and Constructing Models with Multi-Agent Languages* **S26**
 Instructor - Northwestern University (Cross-Listed as LRN_SCI 372,472) - Avg. Class Size: 30

COMP_SCI 396 - *Communicating Computer Science* **W/S 22, S23, S24, S25**
 Course Designer & Instructor - Northwestern University - Avg. Class Size: 36

COMP_SCI 396 / 496 - *Pedagogical Design in Computer Science* **S25, S26**
 Course Designer & Instructor - Northwestern University Avg. Class Size: 22

COMP_SCI 212 - *Mathematical Foundations of Computer Science* **Fall 2022**
 Instructor - Northwestern University

SESP 310 - *Introduction to Statistical Causal Inference* **Spring 2020**
 Course Designer & Instructor - Northwestern University

SESP 210 - *Intro. to Statistics and Research* **W/F 16, W/F 17, W 19**
 Recitation Instructor & Guest Lecturer - Northwestern University

OTHER TEACHING EXPERIENCES

Searle Fellow, Searle Center at Northwestern University **Fall 2023 - Spring 2024**

Computational Thinking Summer Institute 2.0 at Northwestern University **Summer 2020**
 Co-designer & Lead Instructor - *Mathematics Track*

Computational Thinking Summer Institute at Northwestern University **Summer 2019**
 Co-designer & Lead Instructor - *Physics Track*

Searle Center Teaching Certificate Program at Northwestern University **Sept. 2017 - June 2018**
 Course Design Project: *An Introduction to Computational Thinking for Undergraduates*

Wolfram Language Certified Instructor **as of July 2017**

YouthDigital (now Apex Learning) **March 2015 - August 2015**
 Instructor - *Course: Modding Minecraft - An Introduction to Java*

Richland County Public Library Summer Programs **Summer 2012**
 Instructor - *Course: Web Design for All*

STUDENTS MENTORED

Elaine Cui (NU CS MS) - *Master Thesis Chair* **2027**

Linda Kong (NU CS BS) - *Undergraduate Research Advisor* **Summer 2026**
Alumnae of Northwestern University Undergraduate Research Grant

Dawei Xei (NU CS PhD) - *Qualifying Exam Committee* **2026**

Rongwei Peng (NU CS MS) - *Masters Thesis Committee* **2026**

Diana Whealan (NU CS MS) - *Masters Thesis Committee* **2026**

Anya Bardach (NU CS MS) - *Masters Thesis Committee* **2025**

Mark Li (NU CS MS) - *Masters Thesis Committee* **2025**

Jack Carroll (NU CS BS) - *CS Education Research - NU SURG* **Summer 2024**
Weinberg College of Arts and Sciences Summer Research Grant

Maryam Hedayati (NU CS+LS PhD) - *Graduate CS Teaching Mentee* **Winter 2023**

TEXTBOOKS

Wilensky, U., Rand, W., & Bain, C. (in prep). *An introduction to agent-based modeling: modeling natural, social, and engineered complex systems with NetLogo (2nd edition)*. Cambridge, MA: MIT press.

PROCEEDINGS

Bain C. & Wilensky, U. (June, 2026). Frame Alignment and Feedback in CT-ified Science Classrooms. Proceedings of the International Conference for the Learning Sciences (ICLS 2026), Irvine, California, U.S.

Wu, S., Peel, A., Bain, C., Horn, M., & Wilensky, U. (June, 2021). Different Paths, Same Direction: How Teachers Learn Computational Thinking in STEM Practices through Professional Development. CoolThink@ JC, 92. In the proceedings of the International Conference on Computational Thinking Education. Hong Kong, Hong Kong.

Bain, C., Anton, G., Horn, M., & Wilensky, U. (June, 2021). From “Authentic Tools” to Authenticity: Using CT to Enable Exploration in Statistics Classrooms. Proceedings of the International Conference for the Learning Sciences (ICLS), Bocuhm, Germany.

Bain, C., Hershkovitz, A., Dabholkar, S., Horn, M., & Wilensky, U. (April, 2021). Identifying Evidence of Student Engagement in CT via Automated Response Analysis. Submitted to the 2021 Annual Meeting of the National Association of Research in Science Teaching.

Hershkovitz, A., Bain, C., Kelter, J., Horn, M. & Wilensky, U. (April, 2021). Teachers’ Perceptions of the Contribution of Computational Thinking to Science and Math Classrooms. Submitted to the 2021 Annual Meeting of the National Association of Research in Science Teaching (NARST).

Bain, C., Dabholkar, S., & Wilensky, U. (2020). Confronting Frame Alignment in CT Infused STEM Classrooms. CoolThink@ JC, 91. In the proceedings of the International Conference on Computational Thinking Education. Hong Kong, Hong Kong.

Wu, S.P., Peel, A., Bain, C., Anton, G., Horn, M., & Wilensky, U. (2020). Workshops and Co-design Can Help Teachers Integrate Computational Thinking into Their K-12 STEM Classes. CoolThink@ JC, 63. In the proceedings of the International Conference on Computational Thinking Education. Hong Kong, Hong Kong.

Bain, C., Anton, G., Horn, M., & Wilensky, U. (2020). Back to Computational Transparency: Co-design with Teachers to Integrate Computational Thinking in Science Classrooms. Proceedings of the International Conference for the Learning Sciences (ICLS 2020), Nashville, U.S.A.

Martin, K., Bain, C., Swanson, H., Horn, M., Wilensky, U. (2020). Building Blocks: Kids Designing Scientific, Domain-specific, Block-based, Agent-based Microworlds. Proceedings of the International Conference for the Learning Sciences (ICLS 2020), Nashville, USA: ISLS.

Kelter, J., Peel, A., Bain, C., Anton, G., Dabholkar, S., Aslan, U., Horn, M., & Wilensky, U. (2020, May). Seeds of (r)Evolution: Constructionist Co-Design with High School Science Teachers. Proceedings of the International Constructionism Conference. Dublin, Ireland.

Bain, C. & Wilensky, U. (2020, March). Vectors of CT-ification: Integrating Computational Activities in STEM Classrooms. Proceedings of the Special Interest Group on Computer Science Education. Portland, Oregon, U.S.A.

Wu, S., Anton, G., Bain, C., Peel, A., Horn, M., & Wilensky, U. (2020, March). Engage Teachers as Active Co-Designers to Integrate Computational Thinking in STEM Classes. Proceedings of the National Association of Research on Science Teaching. Portland, Oregon, U.S.A.

Bain, C., Anton, G., Horn, M., & Wilensky, U. (2019, October). Building Blocks for Agent-based Modeling Can Scaffold Computational Thinking Engagement in STEM Classrooms. Proceedings of Blocks and Beyond, a VL/HCC Workshop 2019. Memphis, Tennessee, U.S.A.

Martin, K., Wang, E. Q., Bain, C., & Worsley, M. (2019, October). Computationally Augmented Ethnography: Emotion Tracking and Learning in Museum Games. Proceedings of the International Conference on Quantitative Ethnography (pp. 141-153). Springer, Cham. (*Nominated for Best Paper*)

Bain, C. & Wilensky U. (2019). Sorting Out Algorithms: Learning about Complexity through Participatory Simulations. Proceedings of the ACM Special Interest Group on Computer Science Education. Minneapolis, Minnesota, U.S.A.

Martin, K., Wang, E., & Bain, C. (2018). On a method of computational ethnography for museum exhibit engagement analysis. Presented at the Multi-Modal Data Analytics in Education Conference 2018 Conference, Chicago, Illinois. U.S.A.

Swanson, H., Arastoopour Irgens, G., Bain, C., Hall, K., Woods, P., Rogge, C., Horn, M., Wilensky, U. (2018). Characterizing Computational Thinking in high school science. In Kay, J. & Luckin, R. (Eds.), Rethinking Learning in the Digital Age. Making the Learning Sciences Count: Proceedings of the 13th International Conference of the Learning Sciences Vol. 2, 871-878.

Swanson, H., Anton, G., Bain, C., Horn, M., & Wilensky, U. (July 2017). Computational Thinking in the Science Classroom. In Kong, S. C., Sheldon, J., & Li, K. Y. (Eds.), In the proceedings of the *International Conference on Computational Thinking Education 2017*. Vol. 1, 17-22. Hong Kong: The Education University of Hong Kong.

Weintrop, D., Bain, C., & Wilensky, U. (2017). Blocking Progress? Transitioning from Blocks-based to Text-based Programming. Paper presented at AERA 2017. San Antonio, Texas, U.S.A.

Bain, C., Vermeer, W., Anton, G., & Wilensky, U. (2016). The Hackable Name Tag: Recording Interaction Data in Classroom Activities and Beyond. Paper presented at aWEAR 2016: Wearable technologies, knowledge development, and learning, Stanford, California, U.S.A.

Bain, C., Dey, S., Nelakuditi, S., & Choudhury, R. R. (June 2013). "Mitigating Distractions from Smartphones," *Mobile Computing and Communications Review*. Vol. 17 (1), pp. 19-20.

Bain, C. (2013). "The Need for Computer Science Outreach," *ACM: XRDS*. Vol. 20 (2), p. 10.

JOURNAL ARTICLES AND CHAPTERS

Hershkovitz, A., Bain, C., Kelter, J., Peel, A., Wu, S., Horn, M.S., & Wilensky, U. (2023). Contribution of computational thinking to STEM education: High school teachers' perceptions after a professional development program. *Journal of Computers in Mathematics and Science Teaching*, 42(1), 35-65.

Kelter, J., Peel, A., Bain, C., Anton, G., Dabholkar, S., Horn M., & Wilensky, U. (2021). Constructionist Co-design: A Dual Approach to Curriculum and Professional Development. *British Journal of Educational Technology*, 52.3: 1043-1059.

Arastoopour Irgens, G., Dabholkar, S., Bain, C. et al. Modeling and Measuring High School Students' Computational Thinking Practices in Science. *Journal of Science Education and Technology* 29, 137-161 (2020). <https://doi.org/10.1007/s10956-020-09811-1>

Swanson, H., Anton, G., Bain, C., Horn, M., & Wilensky, U. (2019). Introducing and Assessing Computational Thinking in the High School Science Classroom. Chapter in Abelson, H. & Kong, S. (Eds.), *Computational Thinking Education*. Singapore: Springer International Publishing AG.

CONFERENCE PRESENTATIONS

Wu, S.P., Anton, G., Bain, C., Peel, A.N., Horn, M., & Wilensky, U. (2021). Tools and Resources for Integrating Computational Thinking into Your Science Classes. NSTA National Conference 2021, Chicago, Illinois, U.S.A.

Gibson, Z., Anton, G., Vermeer, W., Gomez-Zara, D., Bain, C., Brady, C., DeChurch, L.A., Wilensky, U., & Contractor, N. (2017). Cultivating the conference culture: The role of diversity in interdisciplinary meetings. Presented at the First North American Social Networks (NASN) Conference, Washington, D.C., July 26 - July 30.

Gomez-Zara, D., Vermeer, W., Gibson, Z., Bain, C., Anton, G., Brady, C., DeChurch, L.A., Wilensky, U., & Contractor, N. (2017). The role of brokers in academic network building. Presented at the First North American Social Networks (NASN) Conference, Washington, D.C, U.S.A., July 26 - July 30.

Bain, C., Dey, S., Nelakuditi, S., & Choudhury, R. R. (2013). "YogiPhone: Mitigating Digital Distractions," poster presentation, *ACM HotMobile* at Jekyll Island, Georgia, USA.

CONFERENCE WORKSHOPS

- Bain, C. “Non-Major Coursework Focus Group.” Workshop on Advancing Theory, Research, and Practice for Generative AI in University-Level Computing Education. Evanston, IL, U.S.A. June, 2026.
- Dabholkar, S, Aslan, U., & Bain, C. “Modeling Pedagogy.” NetLogo Conference. Chicago, IL, U.S.A. June, 2026.
- Bain, C. & Anton, G. “Integrating Agent-based Modeling in STEM Classes: From Blocks to Text and Back?” Workshop at ACM Special interest Group on Computer Science Education. Minneapolis, Minnesota, U.S.A. February, 2019.
- Brady, C, Weintrop, D., & Bain, C. “Hacking the Conference Badge”. Workshop at the 2nd Annual International Conference on Computational Social Science. Evanston, Illinois. U.S.A. June, 2016

SERVICE

- Searle Center for Learning and Teaching**, Distinguished Fellow
Northwestern University **September 2026 - present**
- Member**, Program Coordinator Hiring Committee **Summer 2026**
Northwestern University - Office of Residential and Academic Engagement
- Willard Residential College**, Faculty Chair **September 2025 - present**
Northwestern University
- Faculty Advisor** FutureDev, Northwestern University **Fall 2024 - present**
- Member** of School of Engineering Academic Standing Committee **Fall 2023 - present**
- Member** of School of Engineering Academic Appeals Committee **Fall 2023 - present**
- Member** of CS Department Awards Committee **Fall 2022 - present**
- Member** of CS Department Teaching Track Hiring Committee **2023, 2024**
- Point of Contact** for High Schoolers interested in CS Research at NU **Fall 2022 - present**
- Willard Residential College**, Faculty Fellow **September 2021 - August 2025**
Northwestern University
- Reviewer** for Communications of the ACM **2022**
- Reviewer** for the Journal of Teaching and Teacher Education **2022**
- Reviewer** for the ACM Special Interest Group on Computer Science Education **2021, 2020, 2019**
- Reviewer** for FabLearn **2020**
- Submissions Committee** for the Learning Sciences Graduate Student Conference **2019**
- Teaching Consultant** at Northwestern University **Fall 2018 - Spring 2021**
Searle Center for Teaching and Learning
- Willard Residential College**, Assistant Faculty Chair **June 2018 - August 2021**
Northwestern University
- Reviewer** for the ACM Interaction Design and Children Conference **2018**
- Reviewer** for the International Conference on the Learning Sciences **2018**
- Computational Thinking Initiative Volunteer** at Wolfram Research **August 2017 - August 2019**
- Wolfram Language Mentor** **August 2017 - June 2021**
- Wolfram Summer Camp Volunteer** **Summer 2017**
- Finding a Home for CS Education in Schools of Education** **Winter 2017**
Invited Summit Participant
- International Conference on Computational Social Science** **June 2016**
Electronic Registration Co-coordinator

SOFTWARE & CURRICULA

Mickelson, J., Mills, J., Anton, G., Bain, C., Horn, M., Wilensky, U. (2020). Sampling Distributions. [Curricular Unit] Evanston, IL: Center for Connected Learning and Computer-Based Modeling, Northwestern University.

Anton, G., Bain, C., Stone, D., Horn, M., & Wilensky, U. (2020). Analyzing a Global Pandemic with Data Science. [Curricular Unit] Evanston, IL: Center for Connected Learning and Computer-Based Modeling, Northwestern University.

Bain, C. & Bertsche, J. (2017). *Baby Behavior Space*: A NetLogo Web Module for Structured Experiments Center for Connected Learning and Computer-Based Modeling. Northwestern, University, Evanston, IL.

Dabholkar, S., Hall, K., Woods, P., & Bain, C. (2017). From Ecosystems to Speciation (curriculum). Adapted from Novak, M., Horn, M., Brady, C., Wagh, A., & Wilensky, U.: ModelSim - Enabling Modeling and Simulation-Based Science in the Classroom. Center for Connected Learning and Computer-Based Modeling. Northwestern University, Evanston, IL.

Bain, C. and Wilensky, U. (2017). NetLogo GenJam - Duple model. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

Dabholkar, S., Bain, C. & Wilensky, U. (2016). NetLogo GenEvo 1 Genetic Switch model. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

Brockmann, D., Bain, C., Hjorth, A. & Wilensky, U. (2016). NetLogo Kicked Rotator model. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

Brockmann, D., Bain, C. & Wilensky, U. (2016). NetLogo Kicked Rotators LevelSpace model. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

Bain, C. (2016). Parallax E-Badge Conference Registration, Management, Survey, and Visualization Tool. Center for Connected Learning and Computer-Based Modeling. Northwestern University, Evanston, IL.

Bain, C. (2016). *Sorting Out Algorithms*: Learning about Complexity through Participatory Simulations (curriculum). Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

PROFESSIONAL SOCIETY MEMBERSHIPS

Association for Computing Machinery

Institute of Electrical and Electronics Engineers (IEEE)

Omicron Delta Kappa National Leadership Honor Society

Upsilon Pi Epsilon Computing Honors Society

Pi Mu Epsilon Mathematics Honorary Society